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Assignment No: 2

Problem Statement:

Implement all Prompt Engineering approaches — Interview Approach, Chain of Thought (CoT), and Tree of Thought (ToT). Compare and contrast all techniques and analyse their applications. Implement Zero-shot and Few-shot prompting and compare their results.

Objective:

- To understand how prompt design shapes the behaviour and output quality of large language models.
- To implement the Interview Approach, CoT, and ToT prompting techniques using a 120B LLM via Clarifai.
- To compare reasoning styles, strengths, and optimal use cases across all three advanced techniques.
- To implement Zero-shot and Few-shot prompting and evaluate differences in consistency and output format.
- To determine which prompting strategy is best suited for different real-world LLM tasks.

S/W Packages and H/W apparatus used:

- Operating System: Windows/Linux/macOS
- Kernel: Python 3.x
- Tools: Google Colab
- Hardware: Cloud-based GPU via Colab
- Libraries and packages used: openai (Python SDK), Google Colab userdata, Clarifai API (gpt-oss-120b, OpenAI-compatible endpoint)

Theory:

Prompt Engineering is the discipline of structuring natural language instructions to guide LLMs towards desired outputs without modifying model weights. Since LLMs are sensitive to phrasing, token ordering, and contextual framing, well-designed prompts can significantly improve response quality, reasoning accuracy, and format consistency.

Interview Approach:

This technique instructs the model to gather contextual information by asking clarifying questions before providing a response. Rather than answering immediately, the model acts as an interviewer, probing for user-specific constraints, requirements, and preferences. It is particularly effective for personalised recommendations and requirements gathering.

Chain of Thought (CoT):

CoT prompting instructs the model to reason through a problem step by step before arriving at a final answer. Including 'Let's think step by step' or structuring the system prompt to elicit reasoning traces

dramatically improves performance on arithmetic, logic, and multi-step analytical tasks by making the reasoning process explicit and traceable.

Tree of Thought (ToT):

ToT extends CoT by generating multiple parallel reasoning branches, evaluating each branch against criteria, and selecting the optimal path. It mirrors deliberate human decision-making where alternative solutions are considered simultaneously. It is most valuable for strategic planning, open-ended problem solving, and complex trade-off analysis.

Zero-Shot Prompting:

Zero-shot prompting provides only a task instruction with no demonstration examples. The model relies on pre-training knowledge to infer the expected format and logic. It is efficient for simple, well-defined tasks and minimises prompt token usage and API cost.

Few-Shot Prompting:

Few-shot prompting embeds several input-output demonstration examples within the prompt before posing the target query. Through in-context learning — with no gradient updates — the model picks up the desired format, tone, and logic from the examples. It is superior for tasks requiring precise output structure or nuanced domain-specific classification.

Technique Comparison:

- Interview Approach: Iterative and conversational — best for information gathering and personalised outputs.
- Chain of Thought: Linear step-by-step reasoning — best for maths, logic, and multi-step analysis.
- Tree of Thought: Multi-branch evaluation — best for strategic decisions and complex trade-offs.
- Zero-Shot: Instruction-only — best for simple tasks where speed and low token cost matter.
- Few-Shot: Example-guided — best for consistent formatting and domain-specific output control.

Methodology:

- Configure the Clarifai client using the OpenAI-compatible SDK with the gpt-oss-120b model endpoint.
- Define a reusable call_llm(messages, temperature) helper function.
- Interview Approach: Prompt as a tech assistant and request 3 clarifying questions before recommending a laptop.
- Chain of Thought: Instruct the model to solve 25 x 12 step by step in plain text.
- Tree of Thought: Ask the model as a student mentor to generate 3 study strategies, compare them, and select the best.
- Zero-Shot: Provide a restaurant review classification system prompt and the target review; record the output.
- Few-Shot: Add 4 labelled example reviews to the same system prompt before the target review; compare output consistency.
- Analyse all five techniques across reasoning depth, token cost, consistency, and applicability.

Use Cases:

- Use Case 1 - Personalised Customer Support: The Interview Approach enables support bots to collect device type, error details, and usage patterns before generating tailored troubleshooting advice, avoiding generic responses.
- Use Case 2 - Financial Analysis with Traceability: CoT helps LLMs perform multi-step ratio calculations and loan eligibility checks with explicit reasoning steps, reducing hallucination risk in high-stakes numeric domains.
- Use Case 3 - Review Classification at Scale: Few-shot prompting with 3-5 domain examples trains consistent Sentiment-Issue formatted outputs for e-commerce review classification without any model fine-tuning.

Advantages:

- All techniques improve LLM output without requiring model retraining or weight updates.
- CoT explicitly externalises reasoning, making outputs auditable and easier to debug.
- ToT explores multiple solution paths, lowering the risk of a suboptimal single-path answer.
- Few-shot in-context learning adapts instantly to custom output schemas using only a few examples.
- Zero-shot minimises token usage for simple, high-volume classification tasks.

Limitations:

- CoT and ToT substantially increase prompt length and API inference costs.
- The Interview Approach adds latency through multiple conversational turns.
- Poorly chosen few-shot examples can introduce systematic bias in the output distribution.
- ToT quality is bounded by the model's ability to self-evaluate its own reasoning branches.
- Zero-shot accuracy drops sharply on niche tasks not well represented in pre-training.

Applications:

- Contract analysis: CoT guides clause-by-clause legal reasoning with traceable intermediate conclusions.
- Differential diagnosis: ToT generates competing diagnostic hypotheses, evaluates each against symptoms, and recommends the most probable.
- Review moderation dashboards: Few-shot classification enforces consistent Sentiment-Issue format for e-commerce quality monitoring.
- Personalised tutoring: Interview Approach collects learning objectives before generating tailored study plans.
- Code review automation: Zero-shot identifies common bugs and style violations with minimal prompt overhead.

Conclusion:

This assignment implemented and compared five prompt engineering techniques using the 120B gpt-oss-120b model via Clarifai. The Interview Approach demonstrated structured information gathering; CoT produced transparent arithmetic reasoning; ToT generated and evaluated multiple study strategies before selecting the best. Zero-shot restaurant review classification yielded a generic 'Positive - Food' output, while Few-shot produced the more precisely formatted 'Positive - No Issue' aligned with the demonstrated pattern. The experiment confirms that prompt structure is a powerful lever for controlling LLM reasoning depth, output format, and consistency — with each technique offering distinct advantages depending on task complexity and precision requirements.